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Skill 23.1 Exercise 1

The table below stores all order, customer, and product information in a single table, which results in repeated data.

order_id	customer_id	customer_name	customer_address	product_id	product_name	product_price	quantity
101	C01	Alex Kim	123 Oak St	P10	Laptop	999.99	1
102	C01	Alex Kim	123 Oak St	P12	Mouse	25.99	2
103	C02	Jordan Lee	45 Pine Rd	P10	Laptop	999.99	1
104	C01	Alex Kim	123 Oak St	P10	Laptop	999.99	1

Identify at least two pieces of information that are repeated in the table.

Customer name and address (e.g., *Alex Kim, 123 Oak St* appears multiple times)

Product name and price (e.g., *Laptop, 999.99* appears in multiple rows)

How could this data be reorganized into multiple tables to reduce redundancy?

One table for **customers**

One table for **products**

One table for **orders**

What information would belong in each table

Customers Table

customer_id customer_name customer_address

Products Table

product_id product_name product_price

Orders Table

order_id customer_id product_id quantity

Skill 23.2 Exercise 1

Refer to the tables below,

Customers		
customer_id	customer_name	customer_address
C01	Alex Kim	123 Oak St
C02	Jordan Lee	45 Pine Rd

Products		
product_id	product_name	product_price
P10	Laptop	999.99
P12	Mouse	25.99

Orders			
order_id	customer_id	product_id	quantity
101	C01	P10	1
102	C01	P12	2
103	C02	P10	1
104	C01	P10	1

What is the name of the customer who placed order 101?	Alex Kim
What product was ordered in order 102, and what is its price?	Mouse
How many laptops were ordered in total across all orders?	3
Which customer has placed the most orders?	Alex Kim

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Skill 23.3 Exercise 1

Refer to the tables below,

Customers			Products		
customer_id	customer_name	customer_address	product_id	product_name	product_price
C01	Alex Kim	123 Oak St	P10	Laptop	999.99
C02	Jordan Lee	45 Pine Rd	P12	Mouse	25.99

order_id	customer_id	product_id	quantity
101	C01	P10	1
102	C01	P12	2
103	C02	P10	1
104	C01	P10	1

The following causes an error. Explain.

```
result = customers.merge(products)
```

Missing a common column

In the space below, write code that could be used to merge all three tables. The final table should like the one below,

	order_id	customer_id	product_id	quantity	customer_name	product_name	product_price
0	101	C01	P10	1	Alex Kim	Laptop	999.99
1	102	C01	P12	2	Alex Kim	Mouse	25.99
2	103	C02	P10	1	Jordan Lee	Laptop	999.99
3	104	C01	P10	1	Alex Kim	Laptop	999.99

```
full_orders = orders.merge(customers).merge(products)
display(result)
```

Write code to answer the following questions,

What products did Alex Kim order?	<pre>alex_orders = full_orders[full_orders.customer_name == "Alex Kim"]</pre>
What is the total cost of each order (product price $\times$ quantity)?	<pre>full_orders["total_cost"] = full_orders.product_price * full_orders.quantity</pre>
Which customer spent the most money?	<pre>customer_totals = full_orders.groupby("customer_name")["total_cost"].sum()</pre>

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Skill 23.4 Exercise 1

Refer to the tables below,

products				orders					
	id	description	price		id	product_id	customer_id	quantity	timestamp
0	1	thing-a-ma-jig	5	0	1	3	2	1	2017-01-01
1	2	whatcha-ma-call-it	10	1	2	2	2	3	2017-01-01
2	3	doo-hickey	7	2	3	1	3	1	2017-01-01
3	4	gizmo	3	3	4	2	3	2	2016-02-01
				4	5	3	3	3	2017-02-01
				5	6	4	1	2	2017-03-01
				6	7	1	1	1	2017-02-02
				7	8	4	1	1	2017-02-02

In the space below, rename the *id* column in products to *product_id*, then merge the tables. Store the result in *product_orders*

```
products.rename(columns = {"id":"product_id"}, inplace = True)
product_orders = products.merge(orders)
display(customer_orders)
```

Skill 23.5 Exercise 1

Refer to the tables below,

store_a		orders	
item	store_a_inventory	item	store_b_inventory
0	hammer	0	12hammer
1	screwdriver (nan)	1	15nails
2	nails	2	20saw
3	screws (nan)	3	35duct tape
4	saw	4	6 pvc pipe
5	duct tape	5	15rake (nan)
6	wrench (nan)	6	12shovel (nan)
7	pvc pipe	7	54wooden dowels (nan)

Merge the store_a and store_b inventories with an inner merge.

```
store_a.merge(store_b)
```

What data is missing?

screwdriver, screws, wrench, rake, shovel, wooden dowels

Merge the store_a and store_b inventories with an outer merge.

```
store_a_b_outer = store_a.merge(store_b, how='outer')
```

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Which values are nan or None?

store_a_inventory missing data: rake, shovel, wooden dowels (nan or none)

store_b_inventory missing data: screwdriver, screws, wrench (nan or none)

Skill 23.6 Exercise 1

An ice cream parlor and a bakery have decided to merge. The bakery's menu is stored in the DataFrame bakery, and the ice cream parlor's menu is stored in DataFrame ice_cream.

bakery			ice_cream		
	item	price		item	price
0	cookie	2.5	0	scoop of chocolate ice cream	3.0
1	brownie	3.5	1	scoop of vanilla ice cream	2.95
2	slice of cake	4.75	2	scoop of strawberry ice cream	3.05
3	slice of cheesecake	4.75	3	scoop of cookie dough ice cream	3.25
4	slice of pie	5.0			

Create their new menu by concatenating the two DataFrames into a DataFrame called menu.

```
menu = pd.concat([bakery, ice_cream])
```